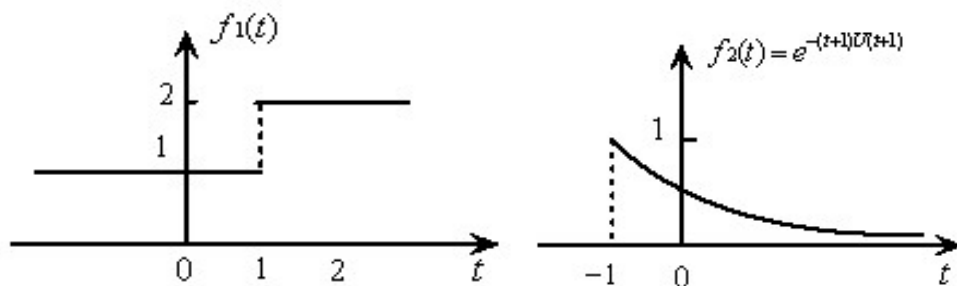


第 5 讲作业 参考答案

5-1 求 $y(t)=f_1(t)*f_2(t)$, 并画出 $y(t)$ 的波形

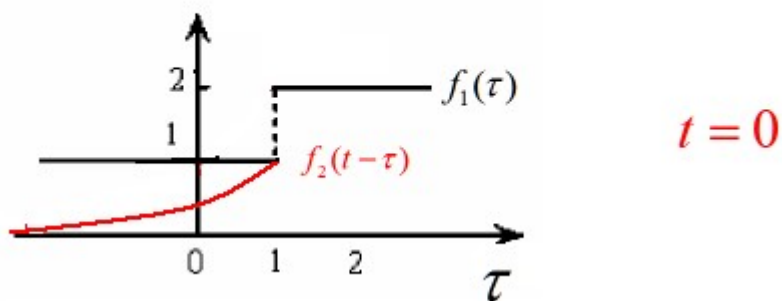


解:

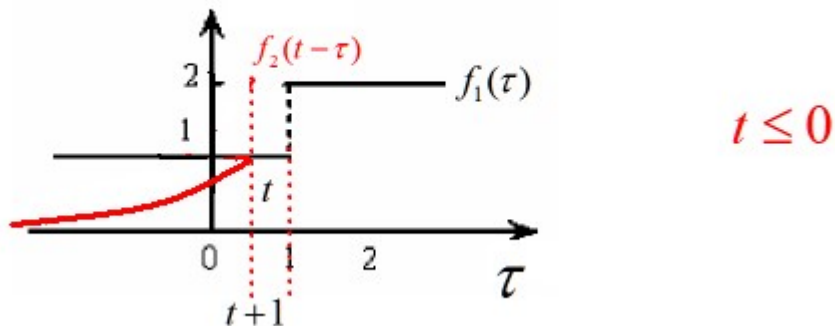
$$f_1(t) = 1 + U(t-1)$$

$$f_2(t) = e^{-(t+1)U(t+1)}$$

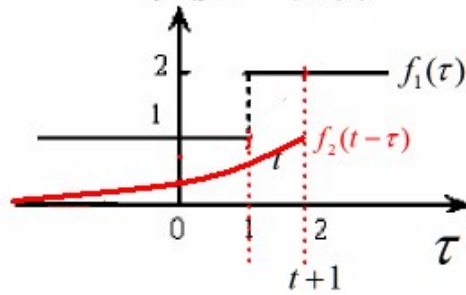
变量替换, 反折



平移, 积分



平移，积分

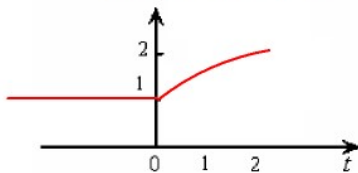


$t > 0$

$$f_1(t) * f_2(t) = \int_{-\infty}^{t+1} e^{-(t-\tau+1)} d\tau = \int_{-\infty}^{t+1} e^{\tau-t-1} d\tau \stackrel{u=\tau-t-1}{=} \int_{-\infty}^0 e^u du = 1$$

$$\begin{aligned} f_1(t) * f_2(t) &= \int_{-\infty}^1 1 \cdot e^{-(t-\tau+1)} d\tau + \int_1^{t+1} 2 \cdot e^{-(t-\tau+1)} d\tau \\ &= \int_{-\infty}^{t+1} 1 \cdot e^{-(t-\tau+1)} d\tau + \int_1^{t+1} 1 \cdot e^{-(t-\tau+1)} d\tau \\ &\stackrel{u=\tau-t-1}{=} \int_{-\infty}^0 1 \cdot e^u du + \int_{-t}^0 1 \cdot e^u du \\ &= 1 + 1 - e^{-t} = 2 - e^{-t} \end{aligned}$$

$f_1(t) * f_2(t)$ 的图形



5-2 证明自相关函数的性质：

- 1) $R_x(0)$ 为自相关函数的最大值： $R_x(0) \geq R_x(\tau)$
- 2) 自相关函数的取值范围： $\mu_x^2 - \sigma_x^2 \leq R_x(\tau) \leq \mu_x^2 + \sigma_x^2$
- 3) $R_x(-\tau) = R_x(\tau)$ $R_x(\tau)$ 是 τ 的偶函数

证明：

- 1) $R_x(0)$ 为 $R_x(\tau)$ 的上界： $R_x(0) \geq R_x(\tau)$ （许瓦尔兹不等式： $E^2(XY) \geq E^2(X)E^2(Y)$ ）

$$\begin{aligned} \text{证明：} R_x^2(\tau) &= E^2(X(t) \cdot X(t-\tau)) \leq E(X^2(t))E(X^2(t-\tau)) = E(X^2(t))E(X^2(t)) \\ &= R_x(0)R_x(0) = R_x^2(0) \end{aligned}$$

- 2) 自相关函数的取值范围： $\mu_x^2 - \sigma_x^2 \leq R_x(\tau) \leq \mu_x^2 + \sigma_x^2$

$$\rho_x(\tau) = \frac{E[(x(t) - \mu_x)(x(t - \tau) - \mu_x)]}{\sigma_x \sigma_x} = \frac{R_x(\tau) - \mu_x^2}{\sigma_x^2}$$

$$\therefore |\rho_x(\tau)| \leq 1$$

$$\therefore \left| \frac{R_x(\tau) - \mu_x^2}{\sigma_x^2} \right| \leq 1$$

$$\therefore \mu_x^2 - \sigma_x^2 \leq R_x(\tau) \leq \mu_x^2 + \sigma_x^2$$

$$3) \quad R_x(-\tau) = R_x(\tau)$$

$R_x(\tau)$ 是 τ 的偶函数

证明:

$$R_x(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t)x(t-\tau)dt = E(X(t).X(t-\tau))$$

$$R_x(-\tau) = E(X(t).X(t+\tau))$$

$$\text{令 } t' = t + \tau$$

则:

$$R_x(-\tau) = E(X(t' - \tau).X(t')) = E(X(t - \tau).X(t)) = E(X(t).X(t - \tau)) = R_x(\tau)$$

5-3 证明自相关函数的性质:

$$1) \text{ 互相关函数的取值范围: } (\mu_x \mu_y - \sigma_x \sigma_y) \leq R_{xy}(\tau) \leq (\mu_x \mu_y + \sigma_x \sigma_y)$$

$$2) \quad R_{xy}(-\tau) = R_{yx}(\tau)$$

证明:

$$1) \text{ 互相关函数的取值范围: } (\mu_x \mu_y - \sigma_x \sigma_y) \leq R_{xy}(\tau) \leq (\mu_x \mu_y + \sigma_x \sigma_y)$$

证明:

$$\begin{aligned} \rho_{xy}(\tau) &= \frac{E[(x(t) - \mu_x)(y(t - \tau) - \mu_y)]}{\sigma_x \sigma_y} = \frac{\lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} (x(t) - \mu_x)(y(t - \tau) - \mu_y) dt}{\sigma_x \sigma_y} \\ &= \frac{R_{xy}(\tau) - \mu_x \mu_y}{\sigma_x \sigma_y} \end{aligned}$$

$$\therefore |\rho_{xy}(\tau)| \leq 1$$

$$\therefore \left| \frac{R_{xy}(\tau) - \mu_x \mu_y}{\sigma_x \sigma_y} \right| \leq 1$$

$$\therefore (\mu_x \mu_y - \sigma_x \sigma_y) \leq R_{xy}(\tau) \leq (\mu_x \mu_y + \sigma_x \sigma_y)$$

$$2) R_{xy}(-\tau) = R_{yx}(\tau)$$

$$\text{证明: } R_{xy}(-\tau) = E(X(t).Y(t+\tau)) = E(Y(t+\tau).X(t))$$

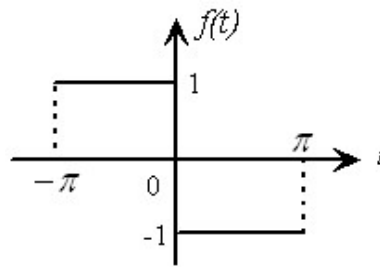
$$\text{令 } t' = t + \tau$$

$$= E(Y(t').X(t' - \tau)) = E(Y(t).X(t - \tau)) = R_{yx}(\tau)$$

第 6 讲作业 参考答案

6-1 如图所示矩形波，请用下列函数近似

$$f(t) = C_1 \sin t + C_2 \sin 2t + \dots + C_n \sin nt$$



任一函数在给定的区间内可以用在此区间的完备正交函数集表示，但若只取函数集中的有限项，或者正交函数集不完备，则只能得到近似的表达式。

$$\begin{aligned} C_n &= \frac{\int_{-\pi}^{\pi} f(t) \sin ntdt}{\int_{-\pi}^{\pi} \sin^2 ntdt} \\ &= \frac{-\int_0^{\pi} \sin ntdt}{\int_0^{\pi} \sin^2 ntdt} \\ &= \frac{\frac{1}{n} \cos nt \Big|_0^{\pi}}{\frac{1}{2} \left[t \Big|_0^{\pi} - \frac{1}{2n} \cos 2nt \Big|_0^{\pi} \right]} = \frac{2}{n\pi} [(-1)^n - 1] \\ C_1 &= -\frac{4}{\pi}, C_2 = 0, C_3 = -\frac{4}{3\pi}, C_4 = 0, \dots \end{aligned}$$

6-2 测 P115 的 2-11

$$\omega = 15, f = \frac{15}{2\pi};$$

$$y(t) = 100 + 109.77 \cos(15t - 10.46)$$

6-3 测 P115 的 2-12

$$(1) a_0 = a_n = 0;$$

$$b_n = \frac{(-1)^{n+1} * 2}{n\pi};$$

$$(2) a_0 = a_n = 0;$$

$$b_n = \frac{[1 - (-1)^n] * 2A}{n\pi}$$

6-4

已知 $f(t) \Leftrightarrow F(j\omega)$ ，求下列信号的傅里叶变换。

$$(1) tf(2t)$$

$$(2) (t-2)f(t)$$

$$(3) (t-2)f(-2t)$$

$$(4) t \frac{df(t)}{dt}$$

$$(5) f(1-t)$$

$$(6) (1-t)f(1-t)$$

$$(7) f(2t-5)$$

$$(8) tU(t)$$

解：(1)

$$\because F[f(2t)] = \int_{-\infty}^{\infty} f(2t)e^{-j\omega t} dt$$

$$\therefore \frac{dF[f(2t)]}{d\omega} = \int_{-\infty}^{\infty} -tjf(2t)e^{-j\omega t} dt = F[-jtf(2t)]$$

$$\therefore \frac{dF[f(2t)]}{d\omega} \Leftrightarrow -jtf(2t) \quad j \frac{dF[f(2t)]}{d\omega} \Leftrightarrow tf(2t)$$

$$\therefore F[f(2t)] = \frac{1}{2} F(j \frac{\omega}{2})$$

$$\therefore tf(2t) \Leftrightarrow jF'(j \frac{\omega}{2})$$

$$(2) (t-2)f(t) = tf(t) - 2f(t) \Leftrightarrow jF'(j\omega) - 2F(j\omega)$$

$$(3) (t-2)f(-2t) = tf(-2t) - 2f(-2t)$$

$$F(tf(-2t) - 2f(-2t)) = jF\left(-j \frac{\omega}{2}\right) - 2F\left(-j \frac{\omega}{2}\right)$$

(4) 因有

$$-jtf(t) \Leftrightarrow \frac{dF(j\omega)}{d\omega}$$

两边同时求导

$$\begin{aligned} -jt \frac{df(t)}{dt} - jf(t) &= \frac{d\left(\frac{dF(j\omega)}{d\omega}\right)}{dt} = \omega j \frac{dF(j\omega)}{d\omega} \\ -t \frac{df(t)}{dt} - f(t) &= \omega \frac{dF(j\omega)}{d\omega} \\ -t \frac{df(t)}{dt} &\Leftrightarrow \omega \frac{dF(j\omega)}{d\omega} + F(j\omega) \end{aligned}$$

所以有：

$$t \frac{df(t)}{dt} \Leftrightarrow -[\omega F'(j\omega) + F(j\omega)]$$

(5)

$$\begin{aligned} F[f(1-t)] &= \int_{-\infty}^{\infty} f(1-t)e^{-j\omega t} dt = \int_{u=1-t}^{\infty} f(u)e^{-j\omega(1-u)} du \\ &= e^{-j\omega} \int_{-\infty}^{\infty} f(u)e^{j\omega u} du = e^{-j\omega} F(-j\omega) \end{aligned}$$

$$(6) (1-t)f(1-t) = f(1-t) - tf(1-t)$$

$$\Leftrightarrow F(-j\omega)e^{-j\omega} - \left\{ j \frac{d}{d\omega} [F(-j\omega)e^{-j\omega}] \right\} = -jF'(-j\omega)e^{-j\omega}$$

$$\begin{aligned} F[f(1-t)] &= \int_{-\infty}^{\infty} f(1-t)e^{-j\omega t} dt = \int_{u=1-t}^{\infty} f(u)e^{-j\omega(1-u)} du \\ &= e^{-j\omega} \int_{-\infty}^{\infty} f(u)e^{j\omega u} du = e^{-j\omega} F(-j\omega) \end{aligned}$$

$$\frac{d(F[f(1-t)])}{d\omega} = \frac{\int_{-\infty}^{\infty} f(1-t)e^{-j\omega t} dt}{d\omega} = \int_{-\infty}^{\infty} -jtf(1-t)e^{-j\omega t} dt = F[-jtf(1-t)]$$

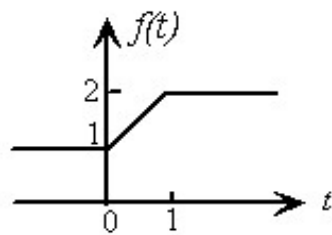
$$\therefore tf(1-t) \Leftrightarrow j \frac{d(F[f(1-t)])}{d\omega} = j \frac{d(e^{-j\omega} F(-j\omega))}{d\omega} = e^{-j\omega} F(-j\omega) + je^{-j\omega} F'(-j\omega)$$

$$\begin{aligned} \therefore F[(1-t)f(1-t)] &= F[f(1-t) - tf(1-t)] \\ &= F[f(1-t)] - F[tf(1-t)] = -je^{-j\omega} F'(-j\omega) \end{aligned}$$

$$(7) f(2t-5) = f\left[2\left(t-\frac{5}{2}\right)\right] \Leftrightarrow \frac{1}{2} F\left(j\frac{\omega}{2}\right) e^{-j\frac{5}{2}\omega}$$

$$(8) tU(t) \Leftrightarrow j \frac{d}{d\omega} \left[\pi\delta(\omega) + \frac{1}{j\omega} \right] = \pi\delta'(\omega) - \frac{1}{\omega^2}$$

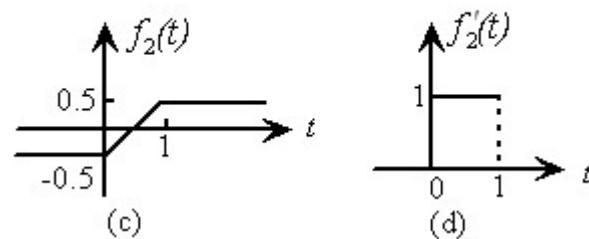
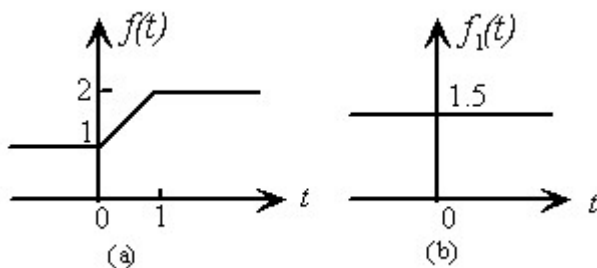
6-5 求如图函数的傅里叶变换。



解：

将 $f(t)$ 分解为 $f_1(t)$ 与 $f_2(t)$ 的叠加。即 $f(t) = f_1(t) + f_2(t)$

如图，

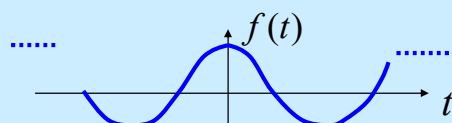


$$F(j\omega) = F_1(j\omega) + F_2(j\omega) = 3\pi\delta(\omega) + \frac{1}{j\omega} \text{Sa}\left(\frac{\omega}{2}\right) e^{-j\frac{1}{2}\omega}$$

也可以直接利用 $t*U(t)$ 和 $U(t)$ 的 Fourier 变换，不过结果要注意化简：

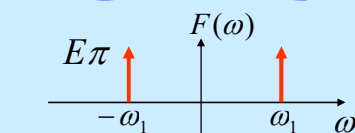
6-6 求周期余弦的功率谱 $f(t) = E \cos(\omega_1 t)$ 功率谱 和自相关

例：求周期余弦的功率谱 $S(\omega)$ 和自相关 $R(\tau)$

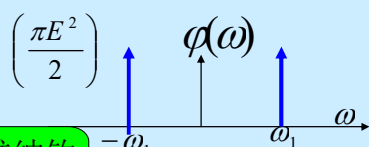


$$f(t) = E \cos \omega_1 t$$

$$= \frac{E}{2} (e^{j\omega_1 t} + e^{-j\omega_1 t})$$



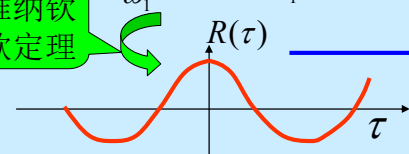
$$F(\omega) = E\pi[\delta(\omega - \omega_1) + \delta(\omega + \omega_1)]$$



$$S(\omega) = 2\pi \sum_{n=-\infty}^{\infty} |F_n|^2 \delta(\omega - n\omega_1)$$

$$S(\omega) = \frac{E^2 \pi}{2} [\delta(\omega - \omega_1) + \delta(\omega + \omega_1)]$$

维纳钦定理



$$R(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S(\omega) e^{j\omega\tau} d\omega$$

$$= \frac{E^2}{4} [e^{j\omega_1 \tau} + e^{-j\omega_1 \tau}] = \frac{E^2}{2} \cos \omega_1 \tau$$

第七讲作业 参考答案

7-1 已知人的脑电波频率范围 0-45Hz，对其作数字信号处理时，求最大抽样周期 T 。如果 $T=5\text{ms}$ 抽样，要使抽样信号通过一理想低通滤波器后，能不失真的恢复原信号，问理想低通滤波器的截止频率 f_c 应满足什么条件？

解：(1)0.011s
(2)45Hz~155Hz

7-2

某地址勘探测试设备给出的发射信号 $x(n) = \delta(n) + \frac{1}{2}\delta(n-1)$ ，接收回波信号 $y(n) = \left(\frac{1}{2}\right)^n u(n)$ ，若地层反射特性的系统函数用 $h(n)$ 表示，且满足 $y(n) = h(n) * x(n)$ 。
求： $h(n)$ ；

解（1）

$$h(0) = y(0)/x(0) = 1$$

$$h(1) = [y(1) - h(0)x(1)]/x(0) = \frac{1}{2} - \frac{1}{2} = 0$$

$$h(2) = \left[y(2) - \underbrace{h(0)x(2)}_{=0} - h(1)x(1) \right] / x(0) = \left(\frac{1}{2}\right)^2 - 0 = \left(\frac{1}{2}\right)^2$$

$$h(3) = \left[y(3) - \underbrace{h(0)x(3) - h(1)x(2) - h(2)x(1)}_{=0} \right] / x(0) = \left(\frac{1}{2}\right)^3 - \left(\frac{1}{2}\right)^2 \frac{1}{2} = 0$$

...

$$h(n) = \begin{cases} 0 & n \text{ 为奇数} \\ \left(\frac{1}{2}\right)^n & n \text{ 为偶数} \end{cases}$$

解（2）

$$h(n) = \left[y(n) - \sum_{m=0}^{n-1} h(m)x(n-m) \right] / x(0)$$

$$\begin{aligned}
&= \left(\frac{1}{2}\right)^n u(n) - \sum_{m=0}^{n-1} h(m) \left[\delta(n-m) + \frac{1}{2} \delta(n-m-1) \right] \\
&= \left(\frac{1}{2}\right)^n u(n) - \frac{1}{2} h(n-1) \\
h(n) + \frac{1}{2} h(n-1) &= \left(\frac{1}{2}\right)^n u(n) \\
\frac{1}{2} h(n) + \frac{1}{2} \frac{1}{2} h(n-2) &= \frac{1}{2} \left(\frac{1}{2}\right)^{n-1} u(n-1)
\end{aligned}$$

7-4: 测 2-17

$$f_s = 2f_m = 2 \times 8.3 = 16.6 \text{ kHz}$$

$$N = \frac{f_s}{\Delta f} = \frac{16600}{1} = 1.66 \times 10^4$$